

TIME SERIES ANALYSIS OF BIBAIRTITE NETWORKS IN ECONOMICS

**S. Takayama^a, H. Aoyama^b, Y. Fujiwara^c, H. Iyetomi^d,
Y. Ikeda^e, and W. Souma^c**

^aDepartment of Physics, Kyoto University
Kitashirakawa-oiwake-cho, Sakyo-ku, Kyoto-shi, Kyoto, 606-8502, Japan
takayama@gauge.scphys.kyoto-u.ac.jp

^bDepartment of Physics, Kyoto University
Yoshida-nihonmatsu-cho, Sakyo-ku, Kyoto-shi, Kyoto, 606-8501, Japan
hideaki.aoyama@scphys.kyoto-u.ac.jp

^cNiCT/ATR CIS Applied Network Science Laboratory
2-2-2, Hikaridai, Seika-cho, Soraku-gun, Kyoto, 619-0288, Japan (ATR)
{yfujiwar, souma}@atr.jp

^dDepartment of Physics, Niigata University
8050, Ikarashi 2-no-cho, Nishi-ku, Niigata-shi, Niigata, 950-2181, Japan
hiyetomi@sc.niigata-u.ac.jp

^eHitachi Research Institute, Akihabara UDX Building, 14-1
Soto-kanda 4-chome, Chiyoda-ku, Tokyo, 101-8010, Japan
yuichi.ikeda.ze@hitachi.com

The study of complex networks has attracted many researchers recently. A particular type of complex networks is the bipartite networks, whose nodes are divided into two sets, F and B, and the only connection between two nodes in different sets is allowed. Many systems in both society and nature are naturally modeled as bipartite networks: movie network is consisted of movie actors and costarring movies, metabolic network is consisted of chemical substances and chemical reactions, etc. The bipartite network we studied is credit network [1, 2]. Credit network has firm nodes, bank nodes and weighted links which means how much money and which firms each banks lends to, in other words how much loan and which banks each firms has from.

There are two ways for study of bipartite network: One is one-mode projection, the other is way of study as it is. One-mode projection onto F means a network containing only F-nodes, where two F-nodes are connected when they have one common neighboring B-nodes. In the former way we can use traditional methods for network analysis, while it is necessary to devise methods for preventing the lose of information. In the latter way we must improve traditional methods, because some network indexes are not valid for bipartite network (e.g. cluster coefficient). We chose an appropriate way for our study and introduce new method and definition if needed.

We have time series data of credit network about thirty years not only network about certain year. There are not many methods and applied example for time series analysis of network, because we must definite network indexes or use comparison methods independent with the number of nodes and links. Our purpose is to argue about methods of analyzing bipartite network as it is and time series analysis, and their application so that we can discover characteristic events form this network and may give some kind of suggestion for the future.

Keywords

credit network, bipartite network, complex network, time series analysis

References

- [1] About the relations between financial institutions;
G. Iori, G. De Masi, O. Precup, G. Gabbi and G. Caldarelli, ``A Network Analysis of the Italian Overnight Money Market," *Economic Dynamics and Control*, 2008.
- [2] About the static structure of credit network;
G. De Masi, Y. Fujiwara, M. Gallegati, B. Greenwald and J. E. Stiglitz, ``An Analysis of the Japanese Credit Network," *preprint*, 2008.